Mechanisms of Reproduction

- Explain the mechanisms of reproduction that ensure the continuity of a species, by analysing sexual and asexual methods of reproduction in a variety of organisms, including but not limited to:

- Animals: advantages of external and internal fertilisation

- plants: asexual and sexual reproduction

- fungi: budding, spores

- bacteria: binary fission

- protists: binary fission, budding

Asexual Reproduction:

- Offspring created is identical as it comes from one parent.
- Use of mitosis.
- Individuals have a short lifespan, so for a population or species to survive, genetic material must be passed from one generation to the next.

Types of asexual reproduction in plants –

- Runners → side branches with clumps of leaves and roots which grow on the ground, the roots dig down and establish the plant as its own individual plant, i.e.. Strawberries.
- Bulbs → bulbs are underneath certain plants which allow buds to grow from them and then flourish their own individual plant eg. Daffodil
- Cutting → branch off a tree is cut and stripped down, then is planted again to grow as its own individual plant.
- Spores → Airborne cells that are released from the parent. They are enclosed and developed when the environment is appropriate. E.g., Fungi → Mushrooms.
- Binary Fission → A parent cell divides, resulting in two identical cells, each having the potential to grow to the size of the original cell. E.g., Amoeba, Bacteria, Paramecium.

Asexual Reproduction in Unicellular Life –

- Single-celled organisms such as bacteria reproduce by simply dividing in two by mitosis. The offspring cells are genetically identical to each other, and to the “parent cell”.
- Among the single-celled, eukaryotic protists such as Amoeba & Paramecium species binary fission (splitting in two) is also common but is often more complex than simple mitosis division.
- In single-celled fungi (yeasts) a process called “budding” is very common. This is a form of binary fission in which a new cell is formed as a small “bud” growing on the parent cell. It separates as a new cell and grows to full size. Each budding cycle doubles the population, so a few cells can become millions very quickly.

Asexual Reproduction in Multicellular Life –

- Many multi-cellular organisms are also able to reproduce asexually.
- Even some animals can reproduce asexually. Perhaps the best-known example is the small aquatic animal Hydra. This is a relative of jellyfish & coral animals.
- Hydra can reproduce sexually by releasing eggs or sperm into the water but can also reproduce asexually by a “budding” process. A small out-growth appears on its body and grows into a new little hydra. Eventually this “baby” separates from the parent to live freely as a separate individual.

Asexual Reproduction	
Advantages	**Disadvantages**
→ Asexual reproduction can produce large numbers of offspring quickly, to take advantage of a sudden or temporary increase in some environmental resource such as food.	→ By producing genetically identical offspring, there is less variation in the population. If an environmental change occurs, a low-variation species is at risk of extinction.

Sexual Reproduction:

- o Involves fusion of genetic material which is the combination of two parents to form an offspring causing high genetic variation.
- o Use of meiosis.

Methods of Sexual Reproduction –

- Process of Pollination → insects, birds or wind carry pollen (male sex gamete) from a flower to another. The pollen attaches to the stigma where the fusion of the two gametes causes a seed to grow where the ovules once were > grows into a fruit.

Sexual Reproduction	
Advantages	**Disadvantages**
→ Sexual reproduction produces more variation in a population, by mixing genes in new combinations. This helps a species survive when environments change.	→ Sexual reproduction is more complex, and often takes more time and energy to achieve.

Internal Fertilisation	External Fertilisation
o The union of an egg and a sperm cell internally through sexual reproduction. Advantages – • Increased possibility of gamete union. • Higher birth rates as foetus grows within an organism not outside of. • More selective of mates. Disadvantages – • Time must be sent trying to attain a	o Fertilisation that occurs outside of the organisms. The sperm cell meets the egg cell outside of the body. Advantages – • Generally, many offspring are formed. • More genetic variation. • Easier to fertilise as they don't need to mate. Disadvantages – • Fusion of gametes may not occur.

mate. • Increased energy must be used to fuse gametes.	• Must occur in water. • Decreased chance of fertilisation.